Hayden Bell | Aspiring Quantitative Developer

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EDUCATION

Heriot-Watt University **Sept. 2024 - Current**

Working towards a B.Sc. in Computer Science. (Expected Graduation 2028) *Edinburgh*

- Notable modules include Discrete Maths, Foundations (Lambda Calculus), Data Structures and Algorithms and Computer Systems
- On track for a First-Class Honours degree; First-class marks in all modules listed
- Grade A in Advanced Higher Computer Science, and grade A in Higher Mathematics at Westhill Academy (2024)

SKILLS

- **Languages:** C++, Python (NumPy, pandas, SciPy), C, C#, Assembly, Java, SQL, Bash, R
- **Systems:** Linux, multithreading and concurrent systems, RTOS, ARM7, CAN bus
- **Quantitative:** Probability, statistics, linear algebra, optimisation

TECHNICAL EXPERIENCE

Aero Watt (IMechE UAS Challenge) **Oct. 2026 – Current**

Software Engineer, Software & Electrical Systems *Heriot-Watt University, Edinburgh*

- Developing autonomy and control software for a fully autonomous fixed-wing UAV, built to complete waypoint navigation, payload delivery and efficiency missions in the IMechE UAS Challenge.
- Implementing mission planning in Python using MAVLink and working on flight control in C++ on ArduPilot/PX4"]
- Collaborating across a 35-member team spanning aerodynamics, airframe and electrical sub-teams to meet competition technical specifications.

HW Racing (Formula Student) **Sept. 2024 – Oct. 2026**

Embedded Software Engineer *Heriot-Watt University, Edinburgh*

- Developed the Vehicle Control Unit firmware for a formula-style electric race car in embedded C on an RTOS, structured as concurrent tasks around a vehicle-mode state machine.
- Implemented CAN message handling, fault detection and watchdog recovery for safety-critical operation
- Applied defensive programming throughout, including bounds checking and static analysis.

PROJECTS | https://github.com/HaydenLBell

- **Backtest Overfitting Detector (Python, NumPy, pandas, SciPy, C++):** Building a tool that estimates the probability a trading strategy's backtest results come from overfitting rather than a genuine edge, using Combinatorially Symmetric Cross-Validation and Deflated / Probabilistic Sharpe Ratio.
- **Real-Time 3D Graphics Engine (C++, OpenGL, GLSL):** Built a real-time renderer from scratch with a programmable shader pipeline, lighting, and a free-moving camera driven by matrix transforms.
- **Stealth FPS Game (Unity, C#, HLSL, SQL):** Built a first-person stealth game with a visibility model that scores how exposed the player is by combining each light source's distance attenuation, angle of incidence and occlusion via ray casting, feeding enemy detection in real time.
- **Stock Trading Simulator (Java):** Built a simulated trading platform that allows a user to purchase stocks and ETFs, calculate estimated profits, and view candles.

AWARDS AND CERTIFICATIONS

- **Awards:** Deputy Principal Award , Heriot-Watt University (x2); 35+ National Competitive figure-skating trophies; 50-hour saltire award ('The Ascent'); and Jane Street Puzzle competitor.
- **Certifications:** CS50: Introduction to Computer Science – Harvard University (edX); Linux Fundamentals – HackTheBox; SOC & Jr. Cyber Security Analyst – HackTheBox